HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

GlucoTwin

A Digital Twin for early glucose-spike warning in Type 2 Diabetes



Type 2 Diabetes

EHR + wearables fused

2-hour early warning

Team Has_Shrey | VCE

Diabetes care is still reactive

101 million

people in India live with diabetes (11.4% of the population)

136 million

more live with prediabetes (15.3%), many undiagnosed



The visibility gap

Clinic visits are occasional. Glucose spikes happen daily, at home.

Repeated high glucose drives long-term damage to the heart, kidneys, eyes and nerves. Today, a spike is usually noticed only after it has happened.

Our focus: one condition (Type 2 Diabetes) and one preventable event (glucose above 180 mg/dL).

Source: ICMR-INDIAB study, The Lancet Diabetes & Endocrinology, 2023

Our idea: a Digital Twin of the patient



1. Fuse

Combine the patient's health record with live wearable and glucose-sensor data.



2. Predict

The virtual patient predicts whether glucose will cross 180 mg/dL in the next 2 hours.



3. Warn and act

The doctor sees the risk early, with reasons, and can act before the spike happens.

Adverse event: glucose above 180 mg/dL within 2 hours | Condition: Type 2 Diabetes | Data: fully synthetic

Two data streams, fused

Static stream: health record (EHR)

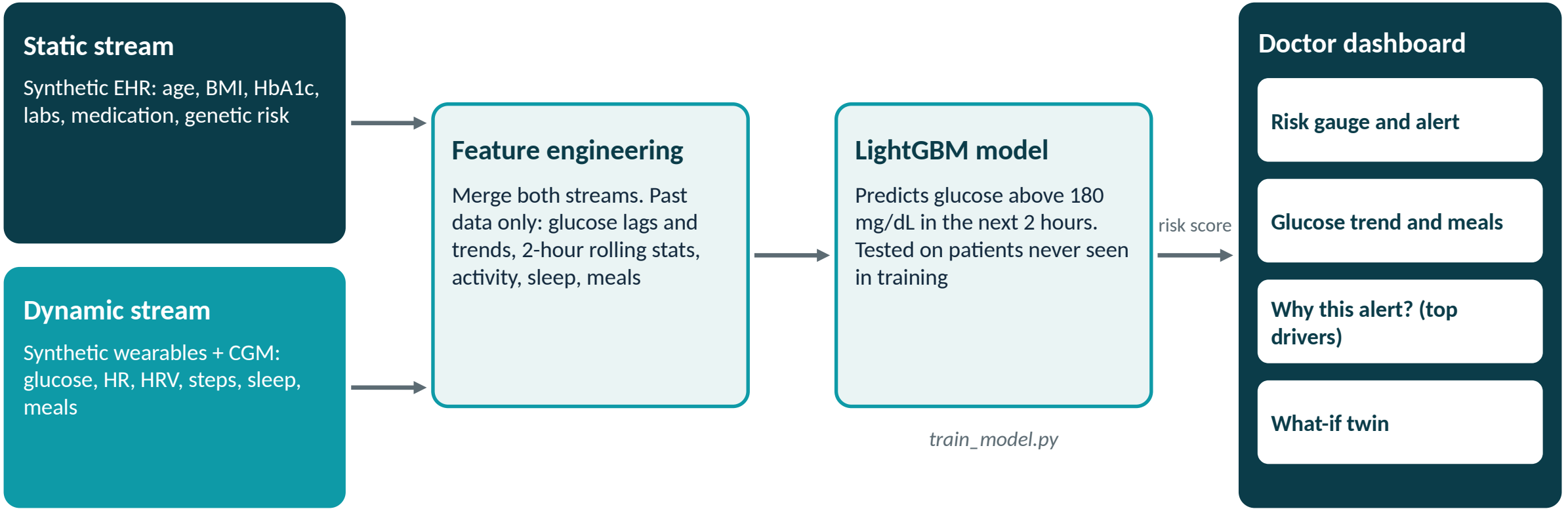
- Age, sex, BMI
- HbA1c, fasting glucose, LDL, kidney function (eGFR)
- Years since diagnosis, hypertension
- Insulin / metformin use
- Family history, genetic risk score

Dynamic stream: wearables and CGM

- Continuous glucose, every 15 minutes
- Heart rate and heart-rate variability (HRV)
- Step count
- Sleep stages
- Meals (carbohydrates logged)

200 virtual patients | 14 days each | about 265,000 readings. No real patient data is used, in line with the challenge's privacy rules (DPDP Act, HIPAA).

System architecture



Runs entirely on a laptop. No special hardware. Synthetic data only.

How the model works

Features from the past only

Glucose 15 min to 2 h ago, trends, 2-hour rolling stats, steps, HRV, sleep and meals. No future data leaks in.

Tested on unseen patients

Split by patient: 136 train, 24 validate, 40 test. The model is judged on people it has never seen.

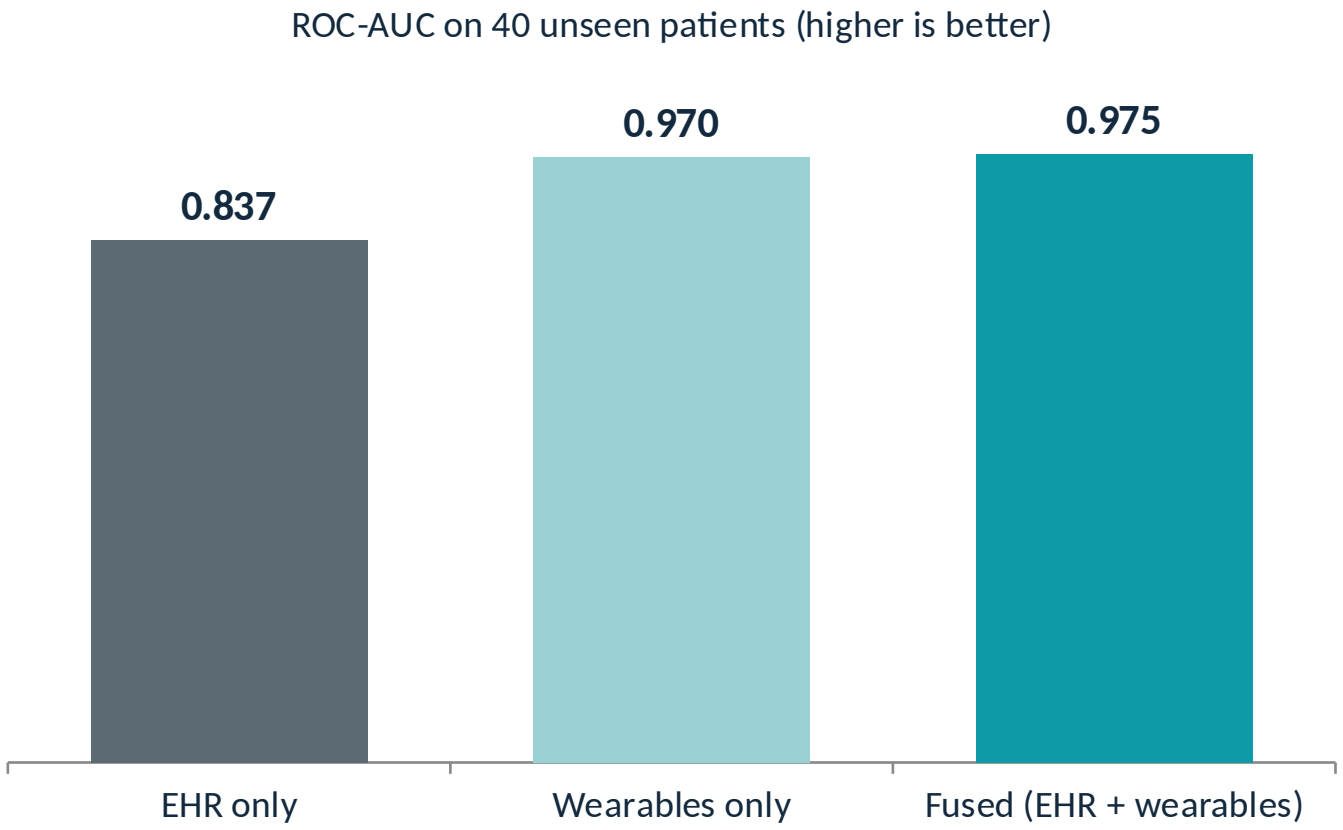
LightGBM classifier

Gradient-boosted decision trees, fast on a laptop and strong on tabular and time-window features.

Explainable alerts

Each alert shows which factors pushed the risk up or down, so the doctor can trust it.

Fusion beats either stream alone



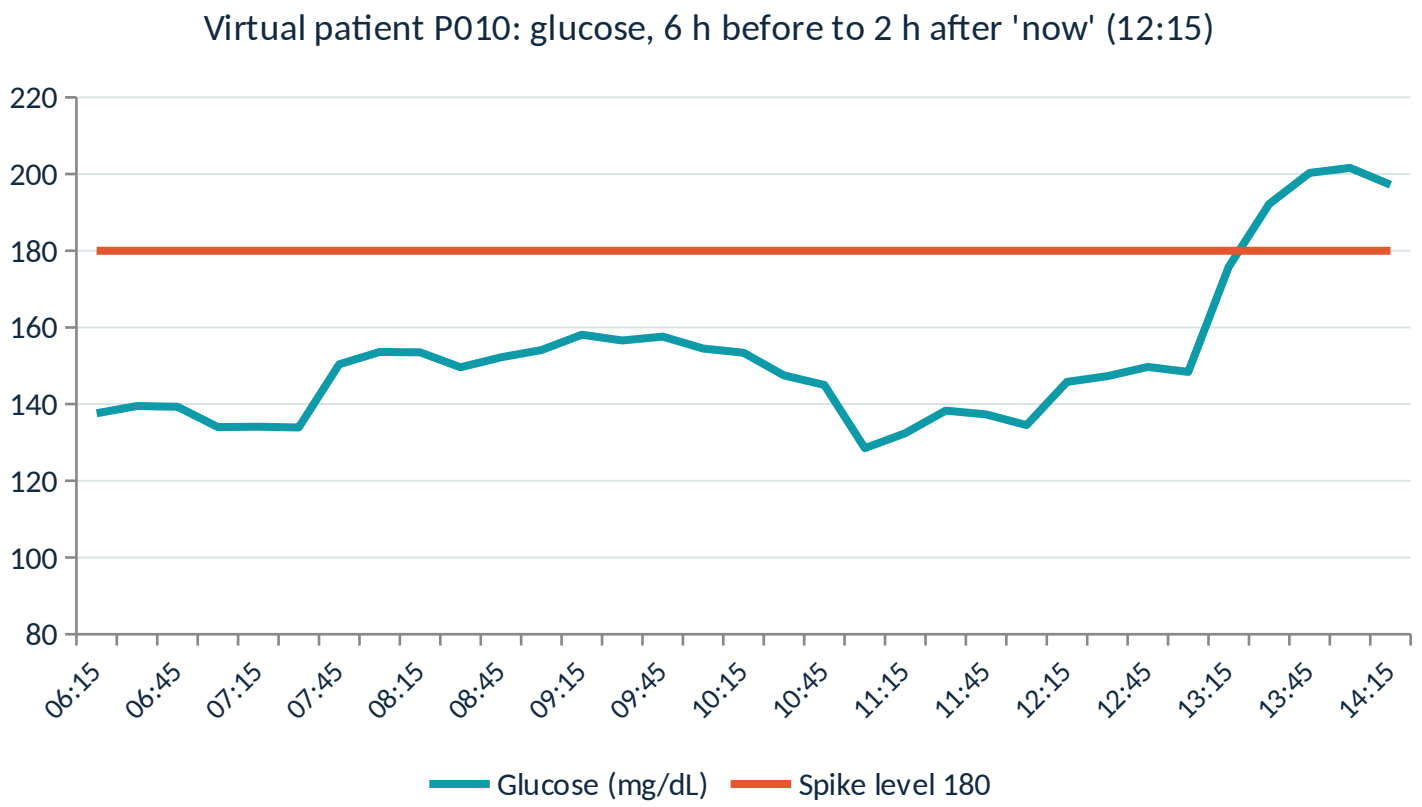
0.87 precision at threshold 0.50: most alerts are real spikes

0.81 recall at threshold 0.50: most spikes are caught early

0.89 recall if the doctor lowers the threshold to 0.30

Synthetic data: these scores show the pipeline works end to end, not clinical accuracy.

The doctor's dashboard



At 12:15: glucose 146 mg/dL

71% risk

HIGH RISK ALERT

Glucose crossed 180 about 75 minutes later

- Risk gauge and adjustable alert threshold
- "Why this alert?" shows the top drivers
- What-if twin: change HbA1c or BMI, see the risk

Chart: real output of our model on an unseen test patient. Live demo follows in the video.

Real-world impact



Doctors

See risk between visits and focus on the patients who need attention first.



Patients

Get an early nudge, such as a short walk or a meal review, before a spike happens.



Health systems

Support remote monitoring at scale and reduce avoidable emergency care.

The same twin design can be extended to hypertension and cardiovascular risk.

Limitations and future scope

Honest limits

- Data is synthetic, so scores show the pipeline works, not clinical accuracy
- Spike behaviour comes from our simulator's rules
- What-if tool covers long-term factors (HbA1c, BMI) only
- Not medical advice; needs clinical validation

Next steps

- Train on open datasets (MIMIC-IV, Synthea records, public CGM data)
- Personalised models per patient
- Live streaming from consumer wearables
- Clinical validation and hospital EHR integration

Thank you

Questions?



Code and documentation

github.com/mshreyas3213-design/Has_Shrey_VCE

Python | pandas | scikit-learn | LightGBM | Streamlit | Plotly | MIT License

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